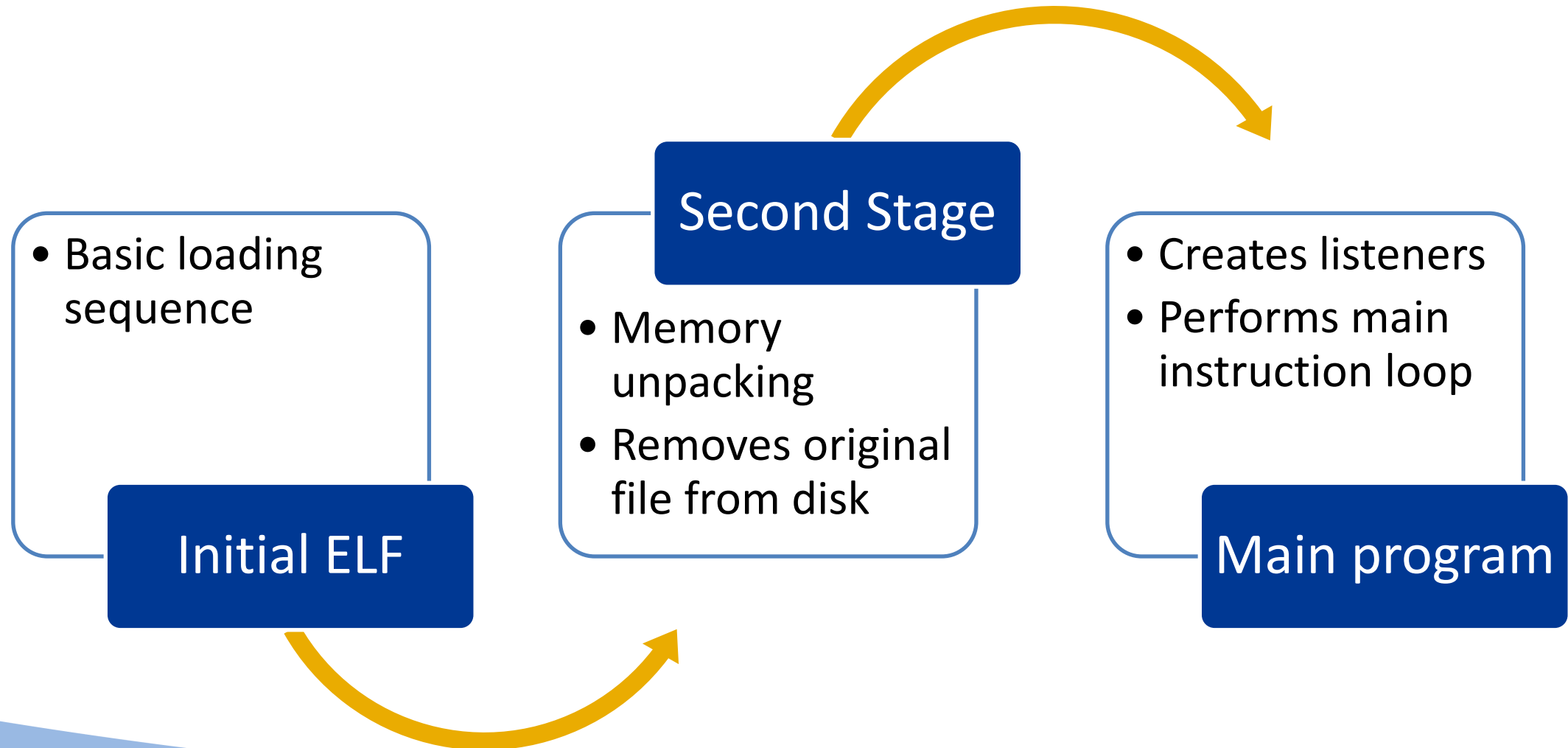


DISMANTLING GAYFEMBOY

Reverse Engineering a 2025 Mirai IoT Botnet
Derivative

High Level Architecture

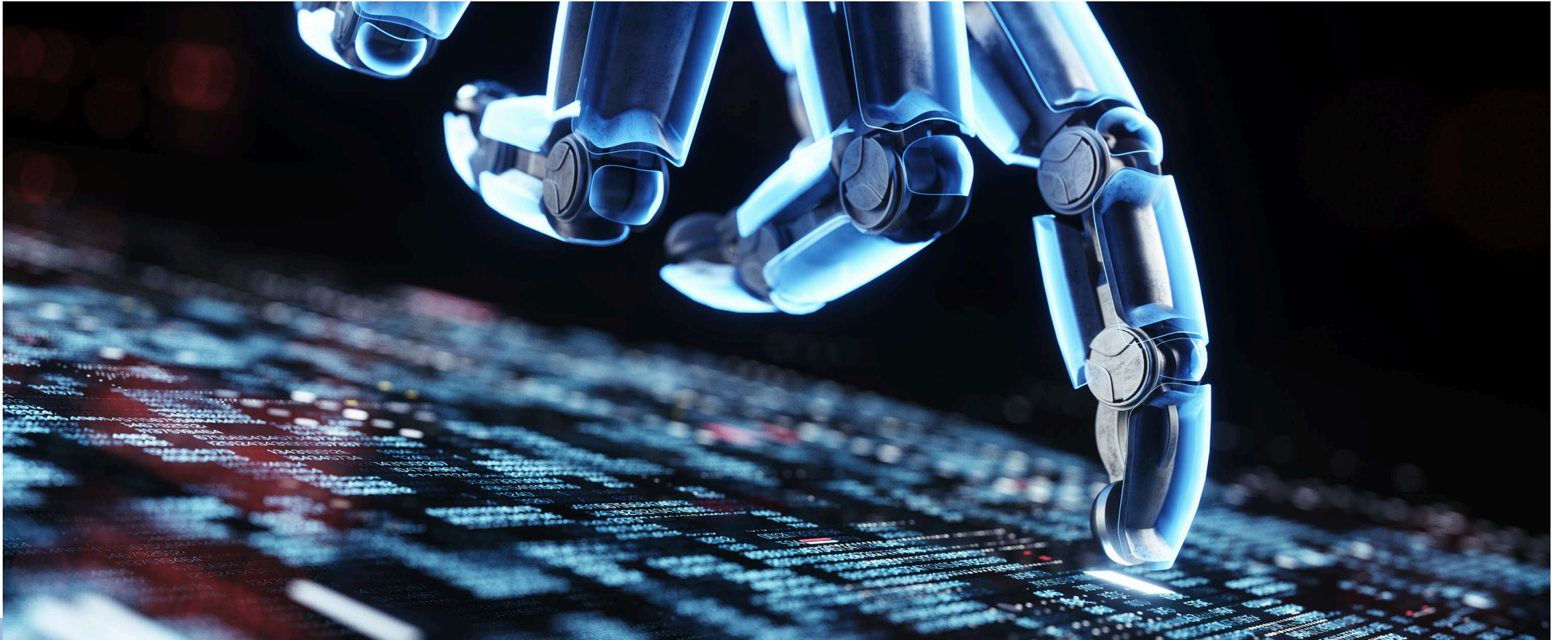


Methodology

- All investigation was performed in an isolated lab environment
 - Main Kali VM
 - RemNux VM (running INetSim)
 - Several static CLI tools (strings, objdump, binwalk)
 - Code analysis through Ghidra
 - Dynamic observations through strace, ltrace, GDB
 - Second and third stages pulled as memory dumps
- 



Live Demo



Key Findings

- Hardcoded C2 domains
- “/proc” enumeration, process-killing, and denylist
- Multi-stage in-memory execution
- Fork-based execution with child liveness signaling

```
                C2 DOMAIN 2: "i-kiss-boys.com" - second hardcoded C2 fallback...
                s_i-kiss-boys.com_0806eda4          XREF[1]:  scanner_build_target_list_from_c...
0806eda4 69 2d 6b      ds      "i-kiss-boys.com"
          69 73 73
          2d 62 6f ...

                C2 DOMAIN 3: "furry-femboys.top" - third hardcoded C2 fallback...
                s_furry-femboys.top_0806edb4        XREF[1]:  scanner_build_target_list_from_c...
0806edb4 66 75 72      ds      "furry-femboys.top"
          72 79 2d
          66 65 6d ...

                C2 DOMAIN 4: "twinkfinder.nl" - fourth hardcoded C2 domain. T...
                s_twinkfinder.nl_0806edc6           XREF[1]:  scanner_build_target_list_from_c...
0806edc6 74 77 69      ds      "twinkfinder.nl"
          6e 6b 66
          69 6e 64 ...
```

```
0806eed5 73 74 72      ds      "strings"
          69 6e 67
          73 00
0806eedd 70 6f 77      ds      "poweroff"
          65 72 6f
          66 66 00
0806eee6 67 63 63      ds      "gcc"
0806eee9 00             ??      00h
0806eeea 67 2b 2b      ds      "g++"
0806eedd 00             ??      00h
0806eeee 6d 6b 6e      ds      "mknod"
          6f 64
0806eef3 00             ??      00h
0806eef4 70 6b 69      ds      "pkill"
          6c 6c
0806eef9 00             ??      00h
0806eefa 6d 6b 66      ds      "mkfifo"
          69 66 6f
0806ef00 00             ??      00h
0806ef01 70 74 72      ds      "ptrace"
          61 63 65 00
0806ef08 73 74 72      ds      "strace"
          61 63 65 00
0806ef0f 63 68 6d      ds      "chmod"
          6f 64 00
0806ef15 6c 73 6f 66    ds      "ls"
0806ef19 00             ??      00h
0806ef1a 62 61 73      ds      "base64"
          65 36 34
0806ef20 00             ??      00h
0806ef21 74 63 70      ds      "tcpdump"
          64 75 6d 70
0806ef28 00             ??      00h
0806ef29 6c 73         ds      "ls"
0806ef2b 00             ??      00h
0806ef2c 69 64         ds      "id"
0806ef2e 00             ??      00h
0806ef2f 77 68 6f      ds      "whoami"
          61 6d 69
0806ef35 00             ??      00h
0806ef36 75 6e 61      ds      "uname"
          6d 65 00
0806ef3c 64 66         ds      "df"
0806ef3e 00             ??      00h
0806ef3f 64 64         ds      "dd"
0806ef41 00             ??      00h
0806ef42 75 70 74      ds      "uptime"
          69 6d 65 00
```

```
syscall_write(1,"twinks :3\n",10);
```

Conclusion

- Modern Mirai is more stealthy, persistent, and dynamic
- Requires both static and dynamic analysis



Questions?

